VNR Vignana Jyothi Institute of Engineering

Computer Science and Engineering | Course Code: CS301

Operating Systems & System Programming - Laboratory Manual & Practical G

Syllabus Focus: CS301 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Kernel Architecture & Process Scheduling

" Syllabus Scope: Microkernel vs Monolithic, process states, PCB, context switching, CPU scheduling: FCFS

Unit 2: Concurrency & Synchronization Primitives

" Syllabus Scope: Critical section problem, Peterson algorithm, hardware test-and-set, semaphores, classi

CS301: Operating Systems & System Programming - Advanced Topics & Assessment

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Deadlock Characterization & Prevention

" Necessary conditions, resource allocation graphs, Banker algorithm for safety and avoidance, detection and r

Unit 4: Memory Hierarchies & Virtual Memory

" Address translation, paging, segmentation, TLB, demand paging, page replacement FIFO, LRU, Optimal, thrashin

Unit 5: Storage Subsystems & File Layouts

" Inode structures, contiguous/linked/indexed file allocation, free-space bitmaps, disk scheduling SSTF, SCAN,

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: POSIX Process Creation fork, exec, wait and IPC via Named Pipes

Lab 2: Multi-Threaded Producer-Consumer Implementation using POSIX Mutexes and Semaphores

Lab 3: Banker Deadlock Avoidance and Safe Sequence Generator in C

Lab 4: Simulation of Page Replacement Algorithms FIFO, LRU, Optimal under Workload Traces

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Abraham Silberschatz, Peter B. Galvin - Operating System Concepts Wiley 10th Ed

* Andrew S. Tanenbaum - Modern Operating Systems Pearson 4th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students